Shaikh Sharik Mohd Arif

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
A.I Abdul Razzak Kalsekar Polytechnic	Class XII (MSBTE)	2020	92.36%
St Mary's Convent High School	Class X (MSBSHSE)	2017	85.20%

PROFESSIONAL SUMMARY

M.Tech Biomedical Engineering student (IIT Hyderabad) with 2 years of industry experience in mechanical design of static equipment. Currently focused on innovating healthcare technology, particularly in the MRI domain, with hands-on experience in ultra-low-field MR system development. I am currently working in Applied Magnetic Resonance lab (aMRIt lab) on different projects mostly related to design, simulation and implementation of optimization algorithms, along with this I am pursuing internship with Siemens Healthineers as an R&D DI-MR intern implementing computational algorithm to improve hardware performance.

COMPETITIVE EXAM QUALIFIED

Qualified

Qualified

Qualified

Qualified

PROFESSIONAL EXPERIENCE

Sr. Engineer - Static Equipment | SS Engineering (Sep 2024 - Jul 2025):

- Design of Reactors, Pressure Vessels and shell & tube heat exchangers
- Preparation of BOM, QAP, GAD, MDS and required calculation excel sheets.
- Ensuring compliance with ASME & TEMA standards.
- Working on ground level with quality dept. and production team for maintaining and handling TPA and production activities.
- Checking of GADs and Design calculation to get it approved.
- Overseeing TQs and providing solution based on available capability.
- Handling design and drafting activities.
- Leading design of equipment based on client's requirement and providing feasible solution.
- Handling meeting with client and TPA to get the job approved for dispatch.
- Cross functional co-ordination with the project, production, QA&QC and purchase department.

Design Engineer | Crystal Industrial Syndicate (Jan 2024 - Aug 2024):

- Design of pressure vessels on PV Elite and ensuring compliance with the required standard.
- Performed FEA validation and calculations on NozzlePRO that are out of WRC bulletin scope.
- To recommend and maintain sheets for Hydrotesting in compliance with ASME standard.

- Preparation and maintenance of calculation sheets and design documents.

- Giving recommendation for PWHT, PFHT and Impact testing based on the code requirement.

- Checking and validating GAD, detailed drawing and design documents and getting it approved.

- Taking part in design meeting at client's end to resolve the current design issues.

- Assisting estimation team by providing preliminary design estimation.

- Cross functional co-ordination with the project, production, QA&QC and purchase department.

- Co-ordinating with client in case of any error in the provided data or due to manufacturing unfeasibility.

- Taking part in project handover meetings. - Assisting quality team for compliance for TPA inspection.

- Correction of design due to manufacturing defecting and getting it approved with client's end.

- Providing tolerance as per the given standard, client's requirement and internal practice.

Sr. CAD Design Engineer | Scantech Laser Pvt Ltd (Jun 2023 - Jan 2024):

- Design of different plans as per client's requirement.

- Handled CNC nesting and sheet optimization.

- Nesting of different components on a standard sheet using metallics software like CncCad and Autonest. Sending proposals and estimation to clients.

- To produce development views of pipes so that it can be taken to produce CNC programming along with it producing and checking programs of pipe cutting and sheet metal cutting.

- Handling of all the Job work provided by the firm ad ensuring that it is on as per given schedule ad timeline.

INTERNSHIPS

Technical Intern R&D (MRI) | Siemens Healthineers | TE-DC-DI-MR (3rd August 2026 -

pursuing till August 2027).

Currently posted in DI-MR research and development department where I am assigned to a

particular problem statement. I am currently responsible for developing a solution that includes the computational compensation of hardware imperfection due to its limited specification. The imperfection is due to it's non-linearities, mechanical and thermal operating limitation. The goal is to prepare an adaptive model that would also take into account the current state of system while leveraging the available sensor's information. With respect to the current research I am trying to implement the hybrid model where under normal condition it can be considered treated as an LTI system where as to account for non linearities TCNN or deep learning based techniques can be implemented.

Summer Internship | Scientific Devices Pvt Ltd (April 2019 - July 2019)

It was a summer internship where I was assigned under production manager. I used look up to the

basic tasks that includes understanding the working principle of different flowmeters and help

with the ongoing inspection tasks of raw materials and assembly. Understanding the calibration

and hydrotesting process of different flowmeters also helping with reporting and documenting

tasks.

PROJECTS

Electromagnetic simulation of implants for MR safety (with HCLTech, ongoing)

I am conducting simulations of the fringe fields generated by an MRI scanner to evaluate their impact on medical implants. When a person moves around the scanner within these fringe fields, the implant may experience deflection from its intended position. Additionally, the varying forces at different locations and orientations within the field can interact with the implant structure. In certain cases, this interaction could potentially impact it's functionality which would may be consisting of different magnetic and ferromagnetic parts.

Shimming simulation and optimization for Ultra-low field system (ongoing)

We are currently developing an ultra-low-field MRI scanner at the aMRIt Lab with a field strength of 52 mT. As part of this effort, I am optimizing the shimming arrangement to achieve a homogenized Bz magnetic field. The target specification is a uniformity of 500 ppm. My work focuses on designing and simulating shimming geometry that minimize field inhomogeneities.

Gradient subsystem design for ultra low field MR systems (ongoing)

I am in the early stages of this project, focusing on studying research papers related to gradient system design for MRI scanners. My current objective is to understand how gradient systems are implemented in conventional MRI setups and how their design requirements differ in the context of ultra-low field systems. This learning process will guide me in optimizing gradient configurations to achieve the desired performance specifications for the 52 mT ultra-low-field MRI scanner under development at the aMRIt Lab.

Eco-friendly Plate Making Machine (Diploma Capstone)

Designed and prototyped a sustainable plate-making system, applying mechanical design principles, material selection, and process optimization. Demonstrated ability to translate theoretical concepts into practical engineering solutions with environmental impact.

Energy Optimization in Domestic Cooking (B.E. Capstone)

Analyzed cooking utensils and thermal efficiency to propose solutions for reducing energy consumption. Applied concepts of heat transfer, fluid mechanics, and system optimization, to analyze the heat losses to the surrounding environment and how much heat is being utilized by the utensils and developed relation between different factors affecting rate of heat transfer analysis and how performance of utensils are compared to each other.

SKILLS

ASME Section VIII Div 1 & 2, Section II, B16.47, B16.5, B16.9, IS 6392, B16.11, B31.3, EEUA guidelines, TEMA basics.

AutoCAD, SolidWorks, Autodesk Inventor, PV Elite.

COMSOL Multiphysics, basic Ansys & MATLAB, Python simulations

Python, Data analytics and Machine learning

RELEVANT COURSES

MRI Principles, Biomedical Imaging, Molecular Imaging, Product Design, Mathematical Physiology, Biomedical Devices

Product Design, Strength of Materials, Fluid Mechanics, Thermal Engineering, CAD, Python Lab, FEA, HVAC, Industrial Electronics,

ADDITIONAL CERTIFICATIONS

IoT Workshop - Pillai's College of Engineering

Data Analytics with Python - IBM

Neuroimaging Workshop - ISMRM 2025

Attended ISMRM Conference 2025